

THE "N" MACHINE EXTRACTION OF ELECTRICAL ENERGY DIRECTLY FROM SPACE

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INTRODUCTION

The extraction of energy directly from space has been suggested as a viable process for the solution of the energy problems of society. The accessibility of this energy has been limited by the necessity for the formulation of new energy paradigms. In the past, energy in space has been suggested by thoughts such as: Orgone, OD, Prana, Bio-cosmic, Neutrino energy sea and so on, but the useful extraction of such energies has always awaited more explicit formulations of these ideas which could suggest the construction of simple practical energy extraction machinery.

VARIABLE INERTIAL MASS

Experiments performed by this author have suggested a picture of space which is perfused with a "fine substance". This concept is one which lies between the ineffability of a space-time construct such as Einsteins and the tangibility of gross matter. This "fine substance" confers inertia on physical objects. The substance of inertia can also be shown to have the property of polarization.

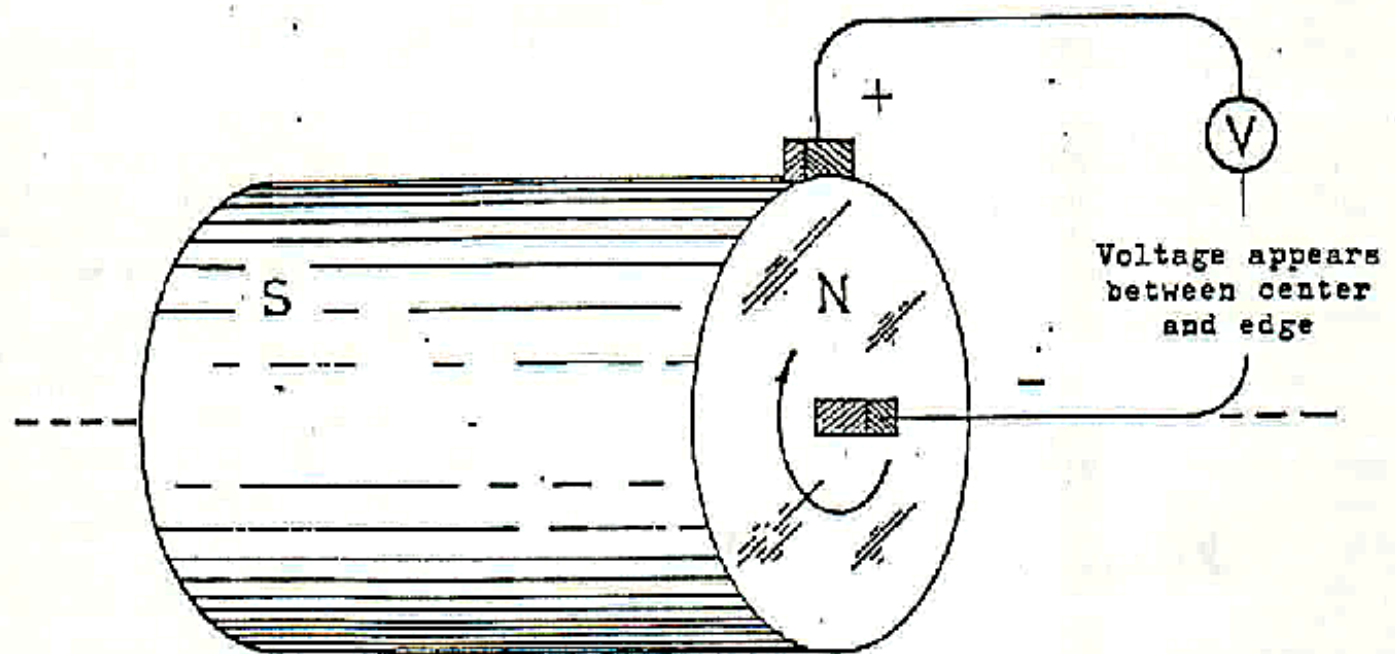
Normally the inertial mass of an object is anisotropic, that is to say, in an inertial measurement performed by applying a force vector to the object and measuring the resulting acceleration, the inertial mass obtained in this measurement would be a constant independent of the direction of the applied force vector.

The important discovery is that the inertial mass of a rotating object becomes polarized and anisotropic. In terms of the real behavior of a rotating object, the inertial mass is found to increase for measurements performed in the direction of the axis of rotation, and perforce the inertial mass is found to be decreased for measurements made in the direction of the plane of rotation. Complete inertial polarization of the rotating object takes place when the inertial mass taken in the direction of the plane of rotation of the test object decreases to zero with increasing rotational speed.

THE "N" EFFECT

The interesting combination would be to combine the effects of inertial and magnetic

The "N" Effect



Rotating Permanent Magnet (Alnico)

FIGURE A

The N Machine

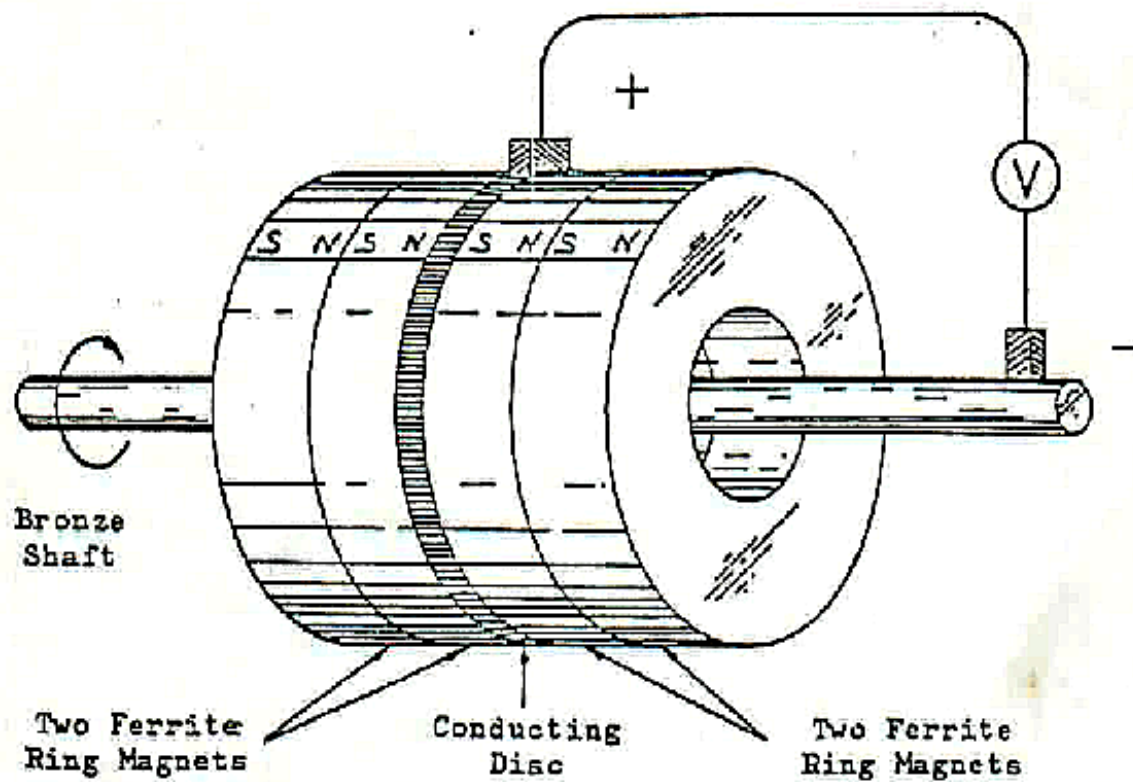


FIGURE B

polarization for the extraction of electrical energy directly from space. With reference to the two diagrams, figures A and B, the "N" effect is demonstrated quite simply. A cylindrical bar magnet of alnico or other magnetized electrical conductor is rotated around an axis passing through the two magnetic poles and perpendicular to the polished pole faces. A cylindrically symmetrical electric field is established within the magnet through rotation. Electrical current is simply extracted by placing the probes or sliding contacts of the appropriate ammeters and voltmeters, one on the axis of rotation and the other on the outer surface of the rotating magnetized conductor.

THE "N" MACHINE

In order to make full use of the current capabilities of an N generator and to accommodate the use of non-conducting ferrite permanent magnets or electromagnets, an "N" Machine is constructed (see Figure B). The "N" machine utilizes a copper or bronze conducting shaft and disc and ferrite "ring" magnets cemented together as shown. A typical machine is constructed with ordinary loudspeaker ring magnets of dimensions: outer diameter 2 7/8 inches, inner diameter 1 1/4 inches and 1/2 inch thickness, two of each epoxy-cemented on either side of a conducting disc 1/8 inch thick. It delivers 30 milliwatts at 3450 r.p.m. The field strength of the magnets as supplied is about 1000 gauss. The current obtainable from the machine is limited only by the resistance of the leads and sliding contacts. Since the aforementioned 30 mw. can be developed across a heavy copper wire shunt of resistance less than .001 ohm, a current in excess of 30 amperes is developed by this simple toy.

ELECTRICAL ENERGY FROM SPACE

Electrical energy developed out of centrifugal extraction of the electrical positive and negative poles from the free energy field of space is supplied in useful

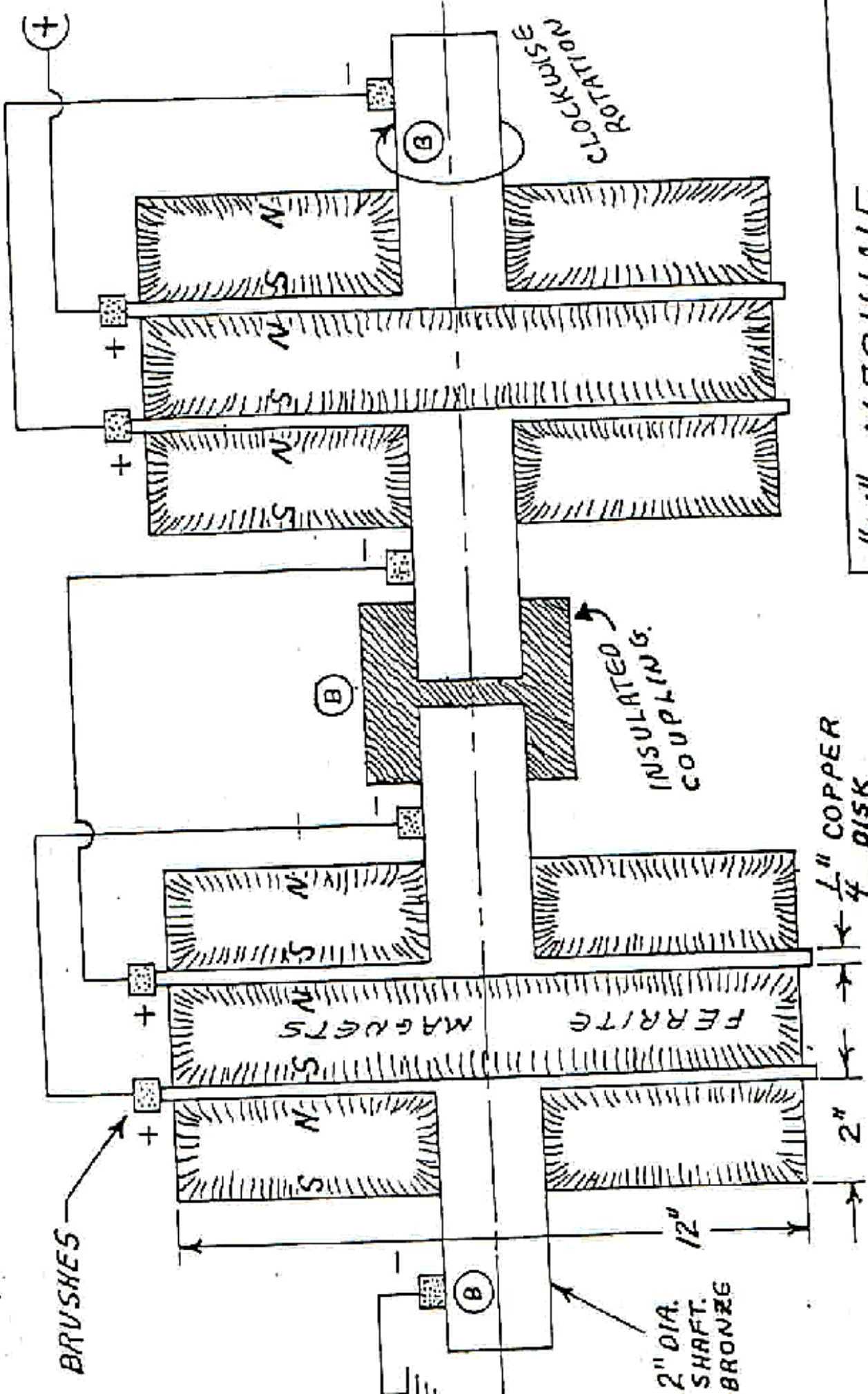
and controllable form from N generators which are scaled in order to supply requirements. Experiments show the voltage polarity depends on the sense of rotation. Output voltage goes directly as speed and magnetic field strength. Geometrically the output voltage increases as the square of the machine radius, r squared.

FARADAY'S EFFECT

It was in the 1830's that Michael Faraday working in the basement of what is now the Royal Technical College in London, discovered that a conducting disc held between the poles of a magnet with the lines of force perpendicular to the surface, would generate current if rotated and the current extracted between the center and the edge of the rotating conducting disc. Conversely if a voltage were applied between the center and the edge of such a disc, it would rotate as a motor. These effects are presently known as the Faraday unipolar dynamo and the Faraday motor respectively. If Faraday had rotated the whole combination, magnets and disc together, he would have discovered, as this author did in 1977, that the voltage output remained constant regardless of whether the disc was rotating independently of the magnets or not. Of course, if Maxwell or Faraday had known of the "N" effect things would have been different. But it is probably true that such a discovery had to wait until the availability of strong lightweight permanent magnets, a development that didn't take place until the 1930's.

NO ROTATIONAL DRAG WITH N GENERATOR

What is important about the N machine is that unlike a conventional generator which exhibits a rotational drag when the current is drawn, an N generator exhibits no such drag. All of the currently used electrical generation rotating machinery has the property of being both motor and generator simultaneously. That is to say, an electrical machine which is used as a generator will



"N" MACHINE

BRONZE AND COPPER WITH
PERMANENT 4000 GAUSS
FERRITE MAGNETS.
12 VOLTS AC @ 600 RPM

3 (B) BALL BEARING LOCATIONS

operate as a motor when excited with the appropriate voltages and currents. With the foregoing in mind, we interpret the situation as follows.

In the conventional electrical power generation system we have an electrical generator coupled to an engine of some kind which supplies mechanical power which is interpreted in accord with present understanding to be converted from mechanical to electrical energy with a conversion efficiency not to exceed 100%. If we were to suppose, however, that the energy obtained was extracted from some heretofore unsuspected property of magnetism; then it is simply apparent that the slowing down of the drive engine is due to the "generator" having the aspect of a motor also. The slowing of the drive engine with electrical load is simply the effect of the motor aspect of the generator energized by the load current. The generator being a motor also elicits a torque output in opposition to the drive engine torque. This is why an engine-generator combination slows down when an electrical load is thrown on the generator.

N MACHINE IS NON-RECIPROCAL

An N generator is only a generator and does not possess the dual aspects of presently used machines. Electrical loading of an N generator produces an internal torque between the conducting electrical disc and the attached ring magnets. However since they are firmly cemented together this torque cannot escape from the machine and load the drive motor or engine.

Thus the N machine is a non-reciprocal machine, which, if a voltage were applied to it in the fashion of motor excitation between the center and the edge of the conducting disc, no motor action could result since the generated torque is constrained within the body of the machine.

DIRECTIONS FOR FUTURE WORK

The discovery of a new physical phenomenon, the N effect, which relates phenomena

of magnetism, inertia, and rotation together in a new machine for the liberation of electrical energy directly from space is a pregnant development of a new age in science which will energize the civilization of the 21st century. Although many ideas may have suggested themselves in the minds of readers of this information, I would like to suggest a few possibilities which have occurred to this author in the time that he has been working and experimenting with N generators of various kinds.

1) CONTROL OF VERY HIGH CURRENTS AT LOW VOLTAGE

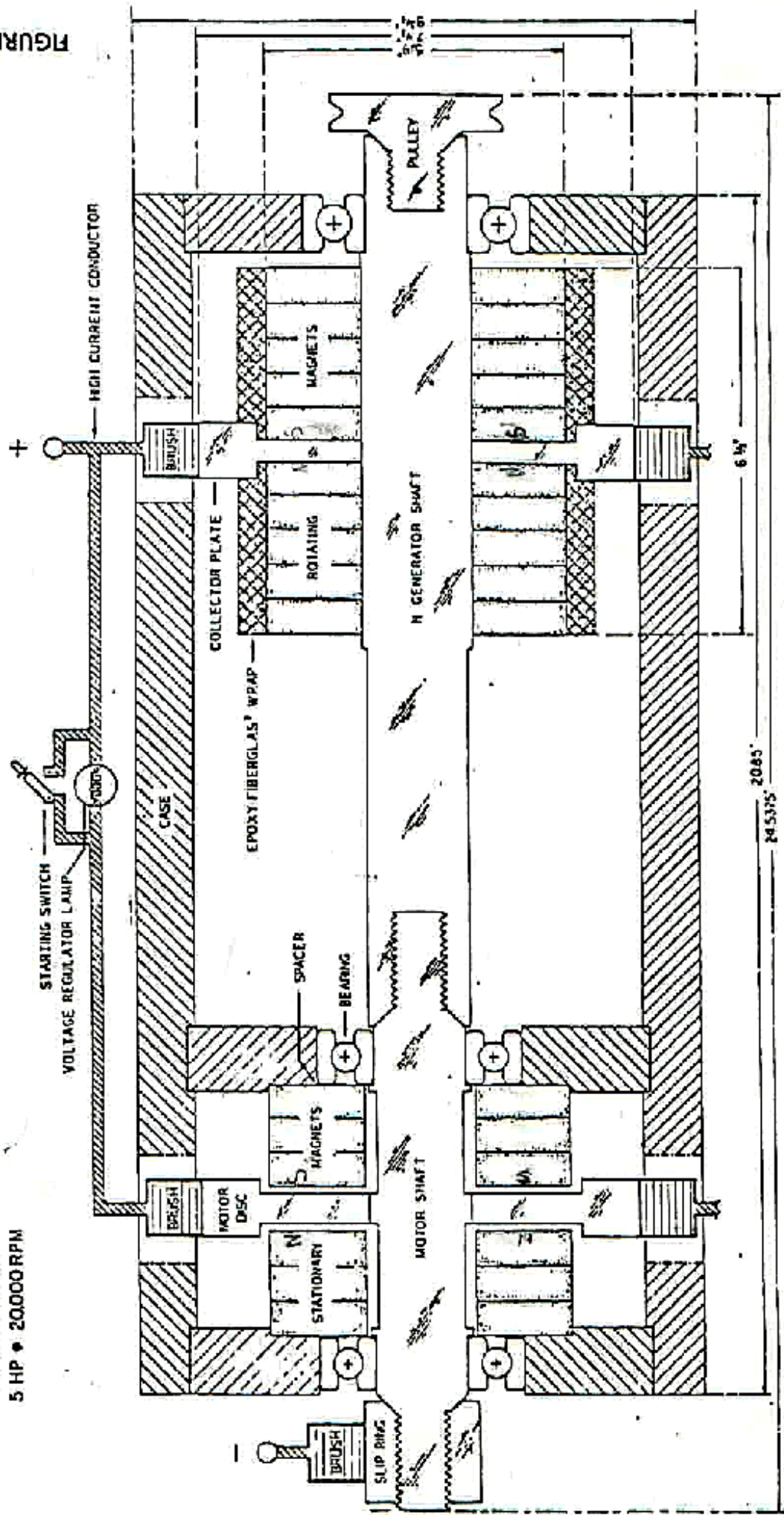
Simple calculations will show the N generator to be characterized as a very high current, low voltage electrical generation machine. For the sizes and rotational speeds normally associated with conventional automotive or electrical traction purposes, it is easy to show that voltage of up to 100 or so v.d.c. can be generated at currents limited only by the brush technology and the machine internal resistance. Standard texts detail methods whereby high currents have been conducted through liquid conducting metal electrodes. In this fashion currents of 50,000 amperes have been conducted from Faraday unipolar generators for the excitation of ultra-high field strength magnets for physical experiments. (Francis Bitter Magnet Laboratory Publications, MIT, Cambridge, Massachusetts 02139).

The important fact about the N generator is that once the appropriate brush technology has been adopted for the ultra-high currents, the control of the voltage becomes very simple. The N generator is constructed as an N machine with the permanent magnets replaced by a pair of electromagnets on either side of the conducting disc. Excitation of the electromagnets can vary the N generator output from zero to full in either polarity. Thus a current of thousands of amperes can be controlled in voltage and polarity by a few amperes or less of excitation current necessary to saturate the electromagnets in

N-1 POWER GENERATION SYSTEM

14 KW • 12VDC

5 HP • 20000RPM



SCALE IN INCHES

FIGURE C

the chosen direction of magnetization.

It is easy to see that a.c. operation is possible if the electromagnets are built of laminations stacked in a cylindrical build with the direction of easy magnetization parallel to the axis of rotation of the machine.

2) SELF CONTAINED POWER GENERATION SYSTEMS

Since the N generator can liberate many times the power needed to overcome bearing friction, windage losses and frictional losses in sliding contacts, the N generator can be combined with an electrical drive motor forming a self-sustaining combination. Reflection will show the appropriate motor for such a purpose is a Faraday motor; a simple copper disc rotor between the poles of strong field magnets. The ultra-low voltage, high current characteristics of this machine combine perfectly with the low voltage, high current output of the N generator. Such a combination, an N generator on a common shaft with a Faraday motor, with the motor excited with a fraction of the generator output regulated through an appropriate series regulator resistor (to prevent machine speed run-away), forms a power generation system. The basic power generation system then consists of a self-sustaining combination of N generator and Faraday motor which provides a mechanical and an electrical output. (See Figure C, The N-I Power Generation System.)

An interesting line of development begins here since once the basic power generation system is constructed, the mechanical output can be used to drive conventional generators—which may be to some advantage since these machines are presently articles of commerce and can deliver higher output voltages than the basic d.c. N generator. The point of all this is that once the free energy is liberated from space and converted into rotational form by a combination N generator-Faraday motor, the resultant energy is directly applicable economically, and with known conventional technology and machines.

3) INERTIAL GUIDANCE

The N generator concept of the direct centrifugal extraction of the electrical poles from the spatial energy field has direct application to the field of inertial guidance. It is not necessary to have sliding contacts if the N generator is to be used to sense de/dt .

Wires can be soldered to the ends of a diameter of an N generator disc and a voltage obtained between the two diametral ends connected together and the center.

The polarity of this voltage will reflect the sense of rotation and its magnitude will be proportional to de/dt . Appropriate Integrators on the output of a three-axis combination will provide all the information necessary for an inertial guidance system replacing cumbersome mechanical gyroscopes spinning at fantastic speeds together with excessively sophisticated and expensive ancillary mechanical and electrical instrumentation.

FREEDOM OF THOUGHT AND ENERGY

The powerful physical principle resulting from the interaction of rotation, inertia, and magnetism, for the liberation of unlimited controlled energy directly from the energy medium of space, the N effect, opens the door to the continued social development of a society freed from the limitation of the present energy conservation paradigm.

Freedom from the limitations imposed by the present formulations, the so-called Laws of Physics, is important since it allows the upward spiraling of free thought which eventually expresses itself in new forms of machinery. In terms of 1979 science, it is an unexpected pleasure to be reminded that the present closed system of equations of electricity, Maxwell's Equations, do not represent all there is to know about electricity and magnetism.

THERMODYNAMIC INTERACTIONS

The Laws of Thermodynamics and the so-called conservation of energy relationships are 150 years old. Of course the discovery of the inertial anisotropy of rotating objects is taken together with new

"N" Machine D.C. Generator

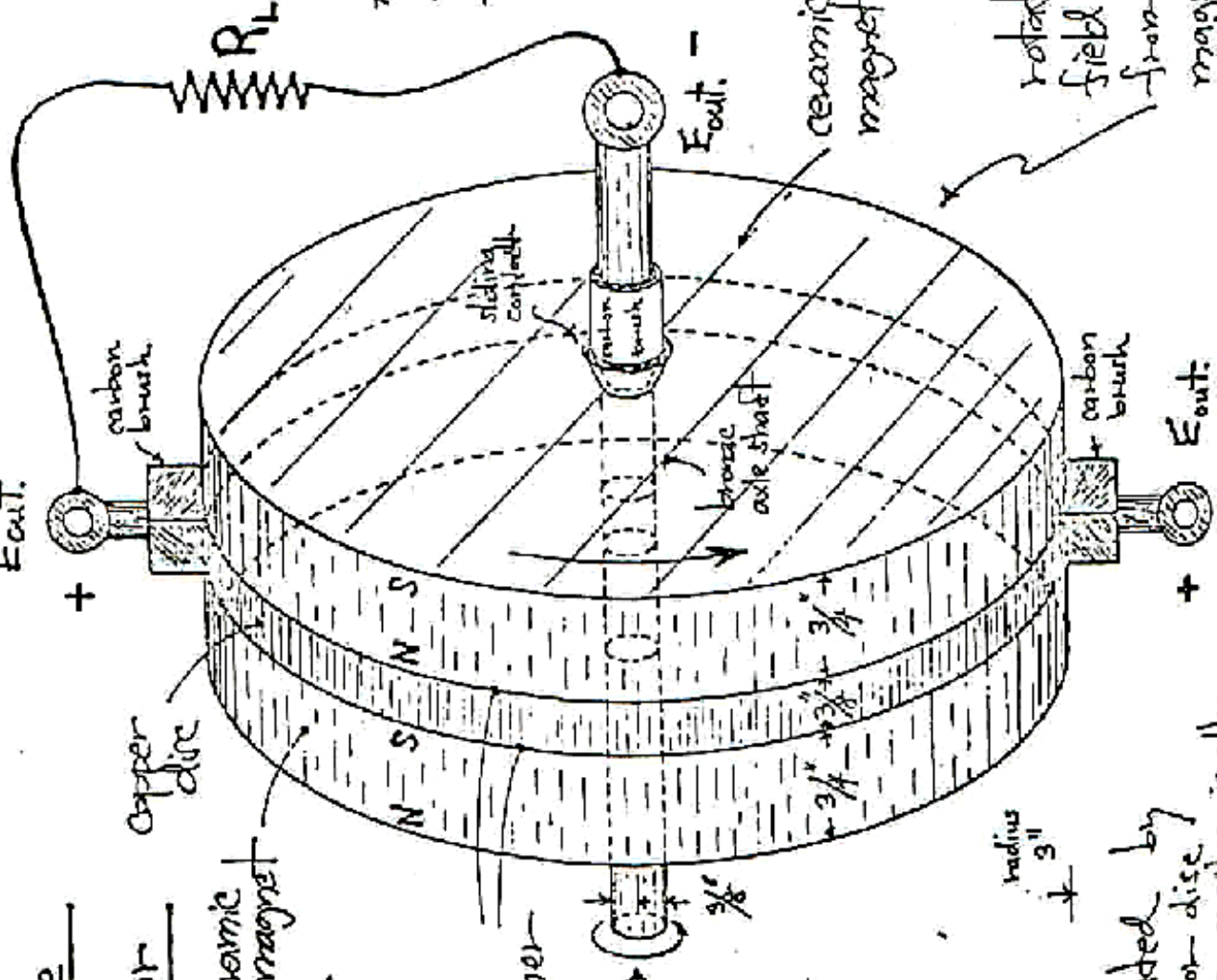
$$E_{out} = \frac{B\omega r^2}{2}$$

composite structure
securely banded together

ω →
constant speed drive

Since this is a free
energy machine the
drive motor must
simply ensure the
driveshaft turn at
constant speed (ω).

A.C. can be generated by
reversing the generator disc
rotation.



rotating magnetic
field must be isolated
from conductive & on
magnetically permeable
materials.

Application of Principle
of Conservation (Force-Energy)
to the generation of
electricity by a ro-
tating polarized
magnetic field.

$$I_L = \frac{E_{out}}{2(R_{int} + R_L)}$$

power output limited
by internal resistance
of copper disc & brushes.

information this author has elicited regarding the elastic collisions of rotating objects impacting on identical non-rotating controls--free energy is liberated in the collision of a rotating object with a non-rotating one. New information such as this imposes new degrees of freedom in the thermodynamic interactions of colliding atoms and will help to explain much of the anomalous new information which is being accumulated in the present search for more "efficient" ways of liberating or extracting energy from Nature.

A thoroughgoing intelligent analysis of the N generator will show that to produce any voltage whatsoever, such a combination of magnets and a conducting disc in rotation as shown, invalidates the physical interpretations of Newton and Einstein, special relativity and general relativity. The loss of these ideas I do not regard as a terribly great tragedy since in their overcoming we shall eventually perfect the anti-gravity spacedrive and will send men to the stars. In this short paper I can only suggest some of these ideas.

OPERATION OF MAGNETRON TUBE

The presently conceived ideas regarding the operation of the magnetron radio frequency transmitting tube can be re-examined in the light of the N effect. In such a tube a rotating disc electronic cloud excites a series of resonant chambers around its periphery at microwave frequencies. The very high power microwave impulses obtained in this way form the basis of Radar transmitters in current usage.

Interpreting the magnetron operation as a higher order property of the Faraday unipolar dynamo, we can suspect we might be able to obtain a.c. excitation of a series resonant L-C circuit connected between any two separate points on the periphery of a rotating N generator conducting disc. For most of us who have spent our lives in the conventional applications of electricity and electrical rotating machinery, it may be enlightening to obtain alternating current in

this way.

What is important is anyone can say a certain formulation or set of ideas is invalid, i.e., the Einstein geometrical interpretation of space. The important thing is what we have to offer in terms of new machinery, i.e., free energy or anti-gravity to substantiate the new ideas.

NEW MACHINERY PROVES NEW IDEA

Experiments performed by this author have obtained 2-3 millivolts a.c. (p-p) generated in this way employing a 1 microfarad capacitor in series with the appropriate inductance to obtain resonance between 100 and 600 cps. In consideration of the utilization of this effect for the generation of megawatt power levels at powerline frequencies (60 cps) the size of the components becomes important since a resonant circuit must be employed in conjunction with an N generator. The L and C elements would have to be fabricated to reach the megawatt power levels with suitably low internal impedance. Such limitations do not appear to assert themselves at the magnetron operating frequencies, so the possibility of the liberation of megawatt power levels of microwave power radiation from an N machine in a UHF cavity system suggests itself.

Consider all the ramifications of the electricity which originally was known in the Galvanic wet cell or the lightning arrester. Now pictures are sent through the "air" (television), sound is recorded (magnetism). Many other things are done. We live in an age where conceptualization of such a development has taken place in many fields. Thus there is some basis of understanding of the possibilities which can result from the evolution of a new basis of understanding. With this in mind I have tried to indicate what some of the thoughts I have had have led me to in consideration of the newly discovered inertial anisotropy of rotating objects and the integration of magnetism and rotation, the N effect.###

Introduction: Beginning in the spring of 1979, the author and his assistants in conjunction with the personnel and facilities of the SUNBURST spiritual and agricultural community constructed a prototype N generator for test and evaluation. After one year of mechanical evolution and refinement, serious testing commenced in August 1980. The important parameter to be considered is the mechanical drag associated with the production of an increment of electrical power which is extracted from space by the machine.

The Experiment: The experimental setup consisted of the N generator machine as diagrammed in figure 2 driven by a drive motor. Specifically the N generator is provided as shown in the illustration with twelve brushes on the axle and also contacting the edge of the disc.

As shown the disc brushes are arranged in three groups of four, arranged on the two sides at top of the disc. The axle brushes are in two groups of six, placed symmetrically on the two sides of the front bearing mount plate. The brushes are 1" x 1" rectangular graphite carbon. The bearings are fitted with an oil lubrication system and the machine is balanced and wound with fiberglass to run without vibration at 6000 r.p.m. The bearing supports and the surroundings of the machine are wood and brass to avoid the proximity of iron or other magnetic material which would induce drag into the machine with the withdrawal of current.

The drive motor is a standard 40 h.p., 440 v.a.c., 60 c.p.s., three phase unit. Motor speed is rated 3485 r.p.m., maximum draw is rated 50.5 amperes at full output. The motor is connected to the N-machine by a double drive belt. The motor pulley diameter is 9 7/8" and the generator pulley diameter is 5 3/4". This gives a step up ratio of 1.72 x the motor speed to rotate the generator at the design speed of 6000 r.p.m.

With initial startup of the machine, the inertial load on the drive motor results in a 200 ampere surge per phase of the drive motor current. After attaining operating speed the drive motor current is 16 amperes at 440 v.a.c. as measured with a Triplatt model 30 ammeter clamp around one of the input phase wires of the motor.

Calculation of the motor power input required to overcome windage and brush friction losses at 6000 r.p.m.:

$$14 \times 440 \times .7 \text{ (p.f.)} \times 1.73 \text{ (pwr. corr 3\phi)} = 7900 \text{ watts} = 10.07 \text{ h.p.}$$

In a number of trial runs the N generator is run up to speed @ 6000 r.p.m. and the brushes are heated by contact with the disc and friction. With the brushes hot their internal voltage drop and series resistance are much reduced over the "cold" condition.

The output of the N generator is regulated by control of the electromagnet. Maximum magnet current is 20 amperes at 200 v.d.c. as provided by a separate power supply and rectifier bridge controlled with a variac operating from the 200 v.a.c. line.

The electrical output of the N machine is measured by passing the current through two 800 ampere rated shunts in parallel. The shunts are Hickok Instruments style 920-221 rated 50 m.v. for 800 amperes passing through them. Two shunts in parallel, mounted on the front bearing support plate absorb the output of the machine. The current is controlled by a large copper knife switch located immediately behind the shunts which controls the current picked up by the 12 brushes riding on the outer edge of the N generator disc.

After the apparatus is started and the N generator attains the 6000 r.p.m. speed the magnet energized and the generator output switch closed to direct the output current into the load consisting of the parallel combination of the 800 ampere shunts. The output of the machine indicated by 50 millivolts appearing across the shunts for every 1600 amperes of current generated by the machine. Voltage is measured by a separate voltage pickup brush located under the front bearing support. The results of six runs off the experiment:

machine speed:	6000 r.p.m.
drive motor current no load	15 amperes
drive motor current increase when N machine loaded	1/2 ampere max.
Voltage output of N generator no load	1.5 volts d.c.
Voltage output of N generator loaded	1.05 v.d.c.
Current output of N generator	7200 amperes
(224 m.v. across shunt @ 50 m.v./1600 amp.)	

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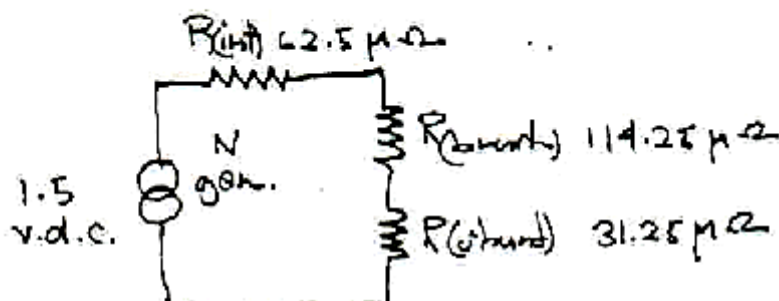
Power output of N machine
Incremental power ratio = 7560
268

7560 watts = 10.03 h.p.
28.2 watts out
watts in

Internal resistance of generator

62.5 micro-ohms

Reduction of the above data gives as the equivalent circuit for the machine:



R (internal) = 62.5 micro-ohms
R (brush) = 114.25 micro-ohms
R (shunt) = 31.25 micro-ohms

The measurement instruments for the voltage and current were:

Voltage: Universal A VOMETER model 8,
1% accuracy

Current: Triplatt model 630-NS, 1% accuracy

Discussion: The figures clearly speak for themselves. During testing when the current output of the N machine was alternatively switched off and on; there was no discernable mechanical change in the condition of the apparatus, i.e. the machinery did not appear to slow in any way and the attitude of the belt and pulley system did not alter showing evidence on any additional mechanical torque being transmitted to the loaded N generator.

The increase in drive motor current was consistent and consisted of a very small discernable movement of the ammeter needle which was exactly at half scale on the 30 ampere range.

The next improvement in the system would be a substitute centrifugal mercury current collected system of conventional design. If we consider such a system the friction losses and heating associated with sliding contacts is eliminated. The electrical resistance can be made very low with such arrangements. Assuming the electrical resistance of the external circuit is made zero through such means then we can expect:

N generator output with zero load resistance = 23,000 amperes
equivalent h.p. = 46.25

It is clear the power output is limited by the internal resistance of the N Generator axle and disc. Earlier experiments with the same machine gave as an internal resistance of 19.2 micro-ohms. Due to the heat liberated in the operation of the experimental machine probably some oxide has formed on the shaft and disc thereby raising the internal resistance on the machine. Liquid metal sliding contacts can be hermetically sealed to eliminate this problem.

Conclusion: Regardless of the later mechanical and electrical improvements which can be made to the construction of the N generator, the point has been proved that the energy which is extracted directly from space by such a machine can be obtained without the drag penalty of the machine constructed along the conventional lines.

* Note: Further evidence of the low drag of the N generator was evident when the drive motor was turned off and the machine allowed to coast down from the normal 6000 rpm operating speed.

Under these conditions opening and closing the current control knife switch to interrupt the flow of the 7200 ampere output current caused no visible effect in the slowing down of the apparatus whether the current was flowing or not. Of course for a generator constructed along conventional lines such a test would result in an immediate arresting of the rotation of the machine as 10 h.p. of electrical load was thrown on the machine.

From the viewpoint of physics, the interesting point is what is the mechanism of the drag of an N machine. I have thought about the problem. The drag only appears when the current is drawn from the N generator disc. No drag is associated simply with the presence or absence of current.

sence of the magnetic field of the electromagnet.

What occurs with the withdrawal of current from the machine is that the magnetic field lines which are normally parallel to the axis of rotation of the machine are distorted into a spiral by the circular magnetic field created in the disc by withdrawal of current. I feel it is the torque interaction of this spiral field with a free space with a non-zero intrinsic magnetic permeability which is resulting in the slight drag which is measured. More precise measurements will clarify this interpretation.

Finally, no change in the current supplied to the electromagnet occurs whether current is being withdrawn from the machine or not. This is consistent with the operation of conventional electrical generators wherein the field excitation current is not altered by withdrawal of current from these machines.

10500 Chms Dr 93108
Simple N-machine made by Bruce dePalma Montecito, Calif. 9-1-6442

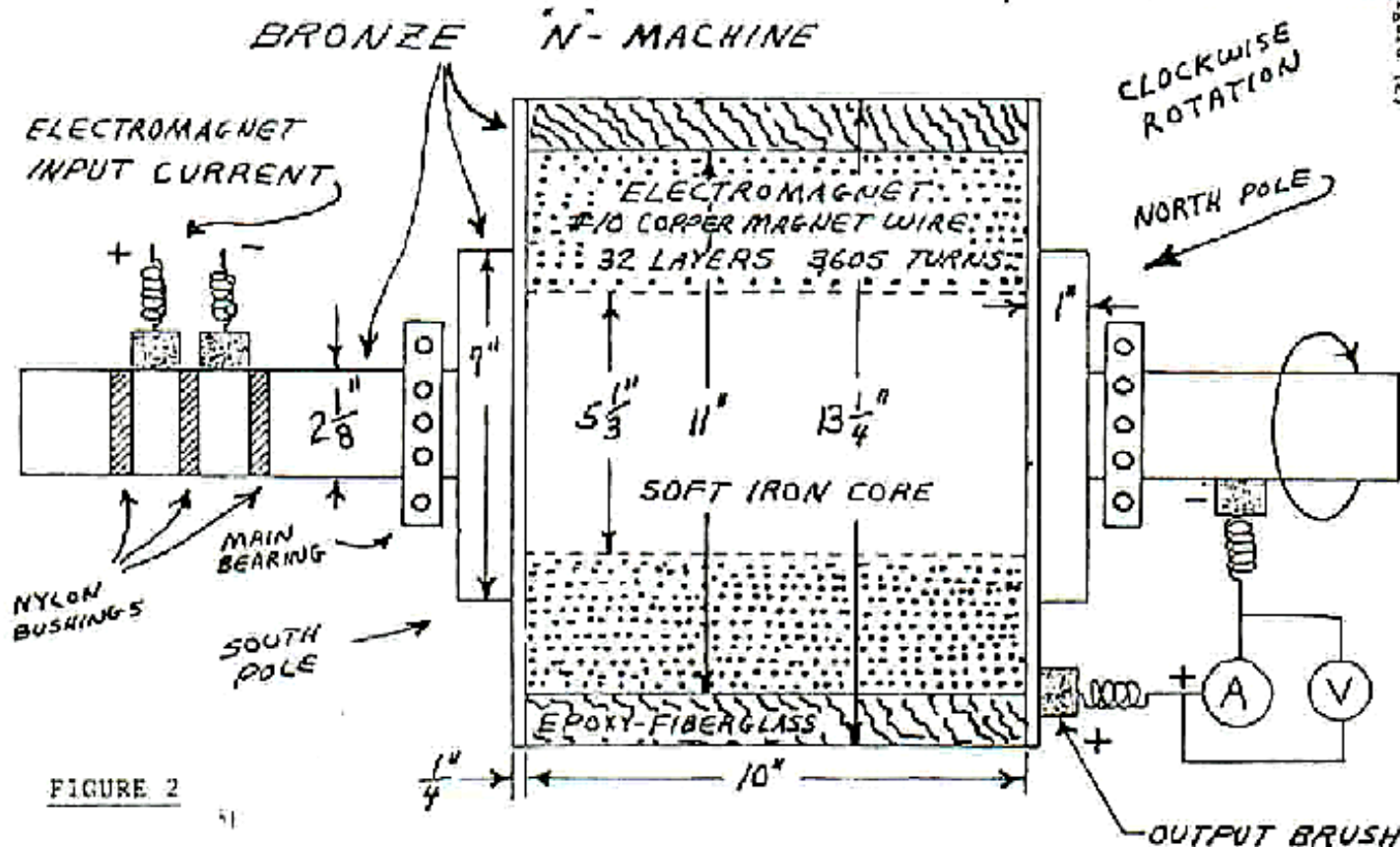
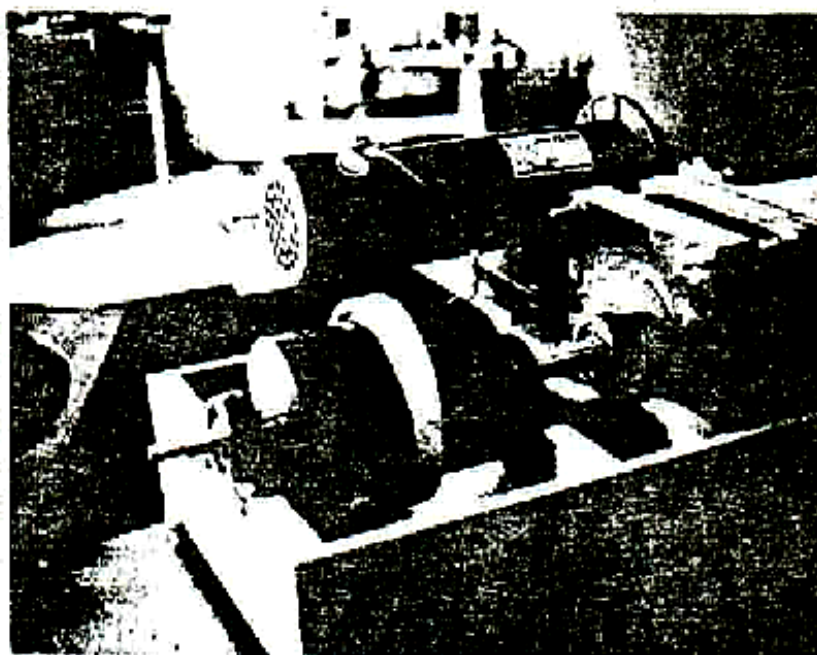
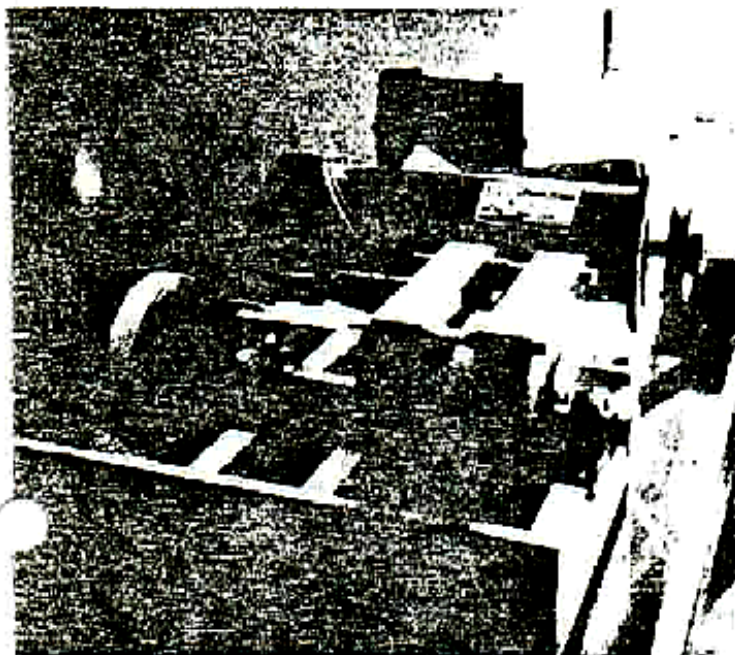


FIGURE 2